Types of Computers & Computer Hardware

Computer Technology

Day 1

Basic Terminology

- Computer
  - A device that accepts input, processes data, stores data, and produces output, all according to a series of stored instructions.
- Hardware
  - Includes the electronic and mechanical devices that process the data; refers to the computer as well as peripheral devices.
- Software
  - A computer program that tells the computer how to perform particular tasks.
- Network
  - Two or more computers and other devices that are connected, for the purpose of sharing data and programs.
- Peripheral devices
  - Used to expand the computer's input, output and storage capabilities.

Basic Terminology

- Input
  - Whatever is put into a computer system.
- Data
  - Refers to the symbols that represent facts, objects, or ideas.
- Information
  - The results of the computer storing data as bits and bytes; the words, numbers, sounds, and graphics.
- Output
  - Consists of the processing results produced by a computer.
- Processing
  - Manipulation of the data in many ways.
- Memory
  - Area of the computer that temporarily holds data waiting to be processed, stored, or output.
- Storage
  - Area of the computer that holds data on a permanent basis when it is not immediately needed for processing.

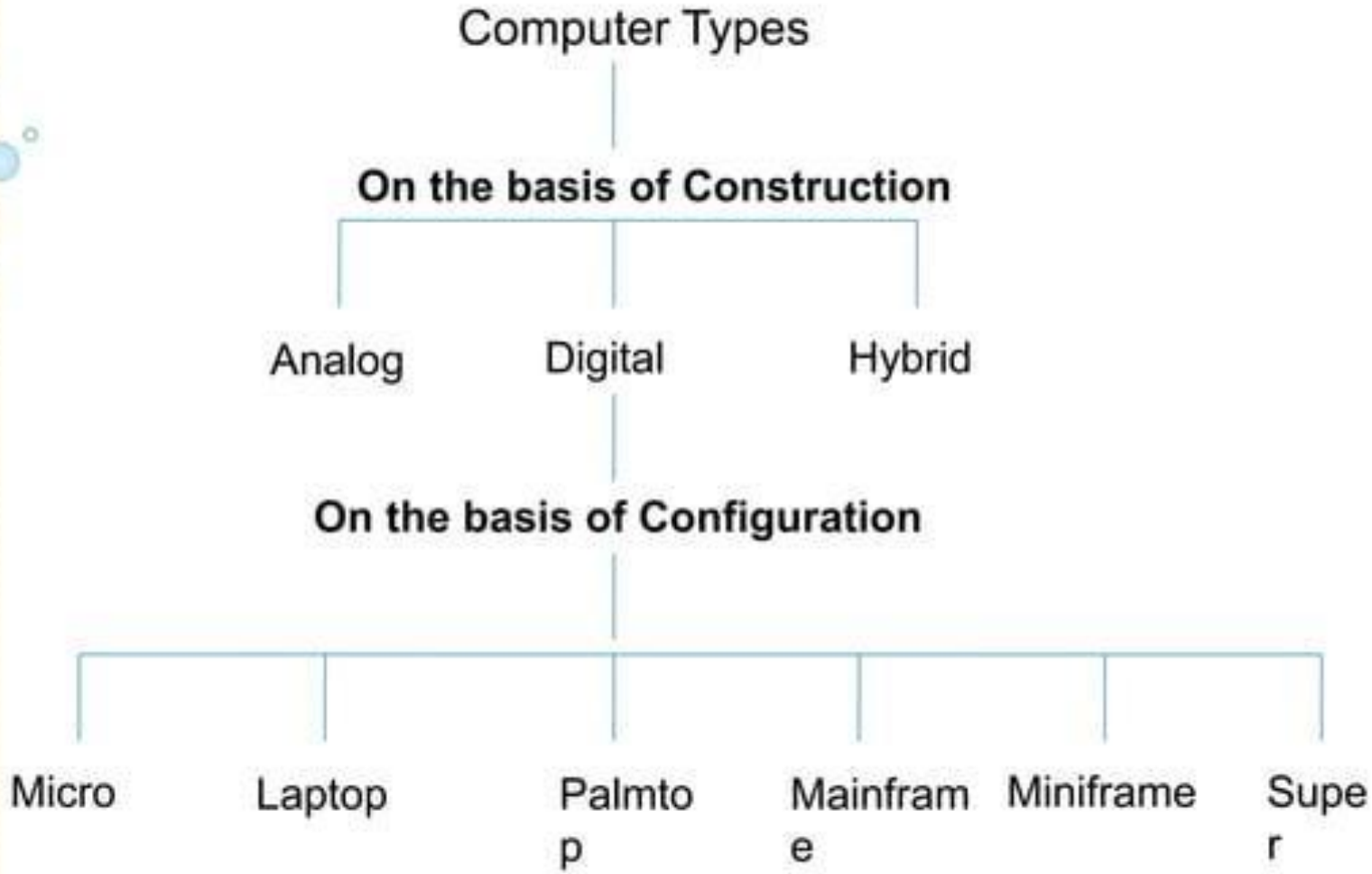
Types of Computers



Computer Types

Computer can be divided into different types on the following basis:

- On the basis of Construction
- On the basis of Configuration


Computer Types
On the basis of Construction
Analog
Digital
Hybrid
On the basis of Configuration
Micro
Laptop
Palmtop
Mainframe
Miniframe
Super

Analog Computer

Analog computers are designed to measure continuous electrical or physical conditions, such as current, voltage, flow, temperature, length, or pressure.

Purpose

- Mostly analog computers are special-purpose computers. Normally they are designed to perform some specific task not multiple tasks.

Analog Computer

- **Key component**
- The key component of the analog computer is the operational amplifier, and the computer's capacity is determined by the number of amplifiers.

Examples

Wrist watch (if non-digital), your car's speedometer, pressure, temperature, and fuel gauges are also considered analog computers.

Analog Computer

Uses

- The output of an analog computer is often an adjustment to the control of a machine; such as, an adjustment to a valve that controls the flow of steam to a turbine generator.
- It is also used in temperature setting to control the ovens for baking.
- They have been widely used in simulating the operation of aircraft, nuclear power plants, and industrial chemical processes.

On The basis of Construction

Digital Computer







Digital Computer

Digital computers are designed to perform daily routine tasks such as writing letters, sending e-mails, performing calculations in an office or creating a database to store large amount of data etc.

Purpose

- Digital computers may be either special or general purpose. Mostly these computers are designed to perform multiple tasks.

Digital Computer

- **Key component today**
- IC (Integrated Circuits)

Examples

Personal Computer, Laptop, Mainframe, Super Computer etc.

Digital Computer

Uses

Word Processing: Word processing is the most common application for personal digital computers.

Accounting - Computers are ideal for keeping payroll records, printing paychecks, billing customers, preparing tax returns, and taking care of many of the other accounting tasks in an organization.

Record keeping - Computers can record information like inventories and personnel files.

Digital Computer

Uses

Industrial Uses - Industrial computers save considerable time and reduce waste by efficiently performing hundreds of industrial tasks.

Science - The research and development applications are the most numerous. Digital computers are being used to do lengthy and complicated mathematical calculations millions of times faster than human beings.

On The basis of Construction

Hybrid Computer





Hybrid Computer

A computer which possesses characteristics of both Analog and Digital computers is called Hybrid computer.

- Some computers combine the functions of both analog and digital computers. These computers are called Hybrid Computers.

Hybrid Computer

Examples

Cement plant, ECG machine, CNG pump

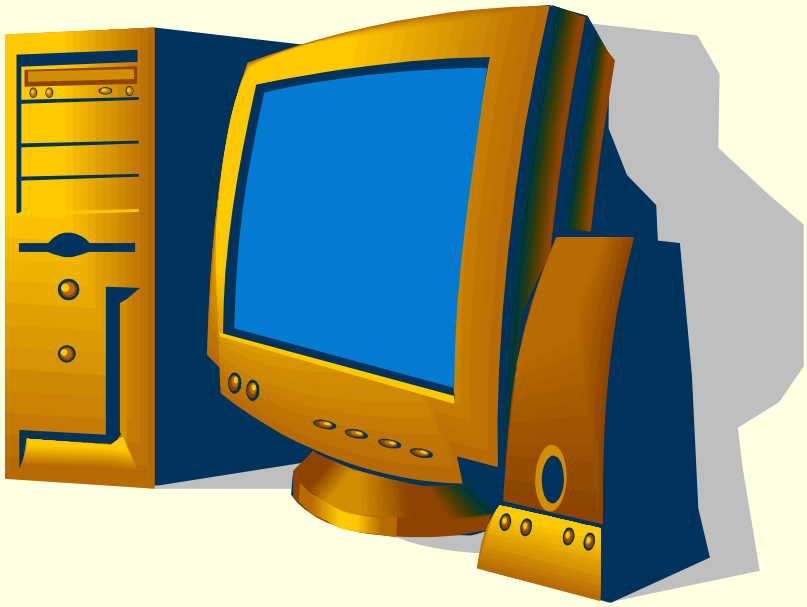


Microcomputer

- A personal computer; designed to meet the computer needs of an individual.
- Provides access to a wide variety of computing applications, such as word processing, photo editing, e-mail, and internet.



Desktop Microcomputer



- A microcomputer that fits on a desk and runs on power from an electrical wall outlet.
- The CPU can be housed in either a vertical or a horizontal case.
- Has separate components (keyboard, mouse, etc.) that are each plugged into the computer.

Laptop Computer



- A portable, compact computer that can run on an electrical wall outlet or a battery unit.
- All components (keyboard, mouse, etc.) are in one compact unit.
- Usually more expensive than a comparable desktop.
- Sometimes called a Notebook.

Workstation

- Powerful desktop computer designed for specialized tasks.
- Can tackle tasks that require a lot of processing speed.
- Can also be an ordinary personal computer attached to a LAN (local area network).

Supercomputer

- A computer that was the fastest in the world at the time it was constructed.
- Can tackle tasks that would not be practical for other computers.
  - Typical uses
    - Breaking codes
    - Modeling weather systems

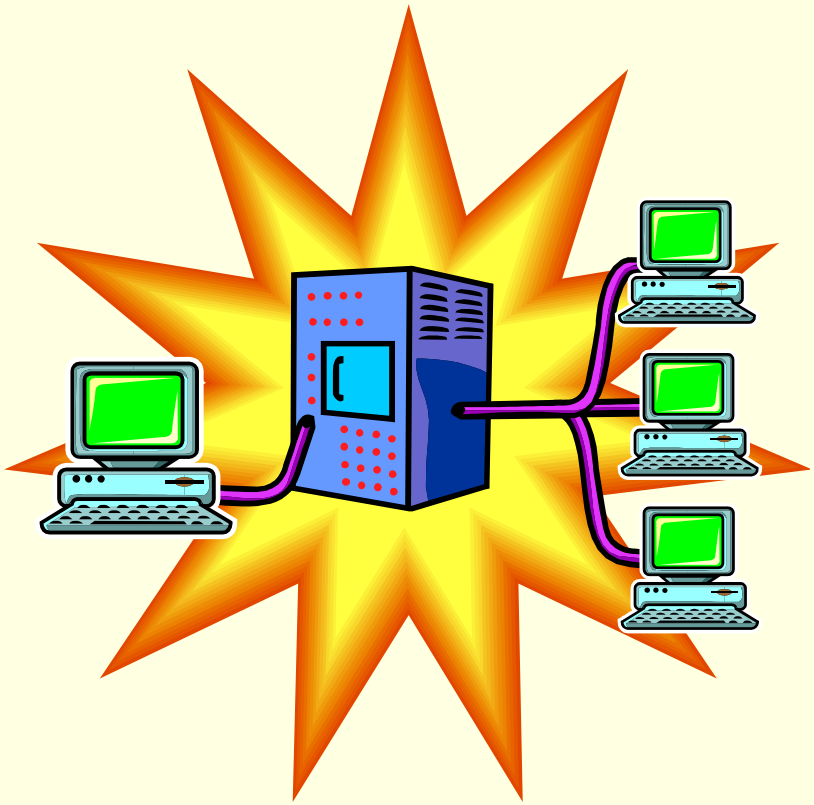
Mainframe



- Large expensive computer capable of simultaneously processing data for hundreds or thousands of users.
- Used to store, manage, and process large amounts of data that need to be reliable, secure, and centralized.
- Usually housed in a closet sized cabinet.

Server

- Purpose is to “serve.”
- A computer that has the purpose of supplying its users with data; usually through the use of a LAN (local area network).



Handheld



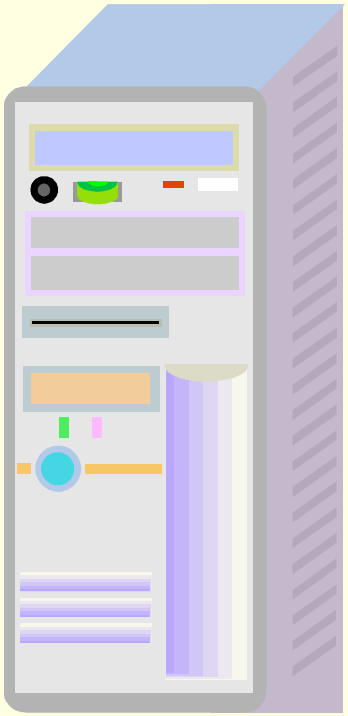
- Also called a PDA (Personal Digital Assistant).
- A computer that fits into a pocket, runs on batteries, and is used while holding the unit in your hand.
- Typically used as an appointment book, address book, calculator, and notepad.
- Can be synchronized with a personal microcomputer as a backup.

Computer Hardware

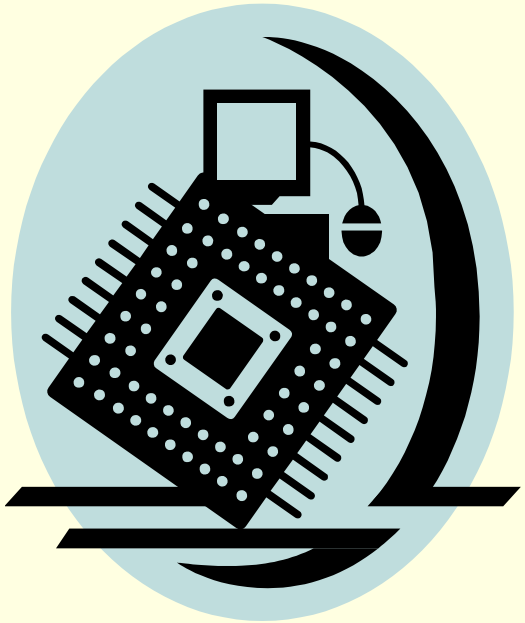


System unit

- Case that holds the power supply, storage devices, and the circuit boards (including the motherboard).



CPU (Central Processing Unit)



- Where the processing in a computer takes place, often called the brain of the computer.

Input Devices

- Units that gather information and transform that information it into a series of electronic signals for the computer.

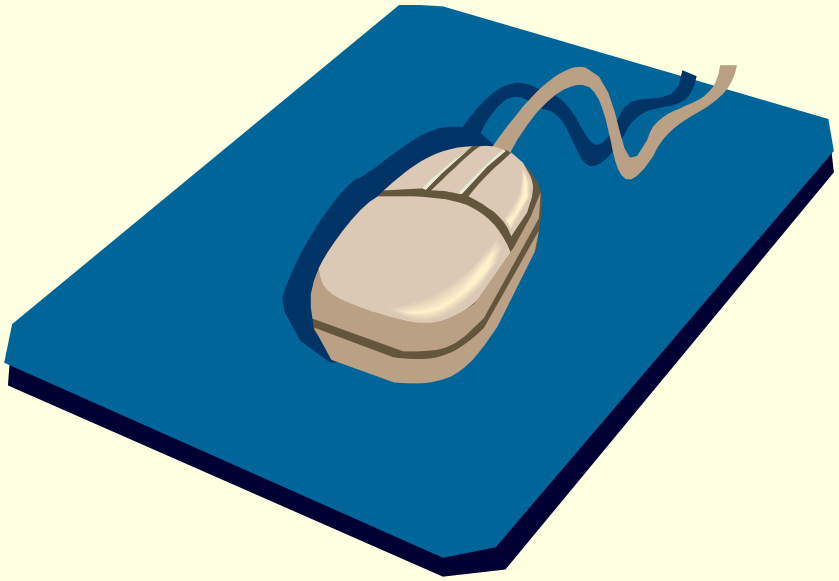
Keyboard



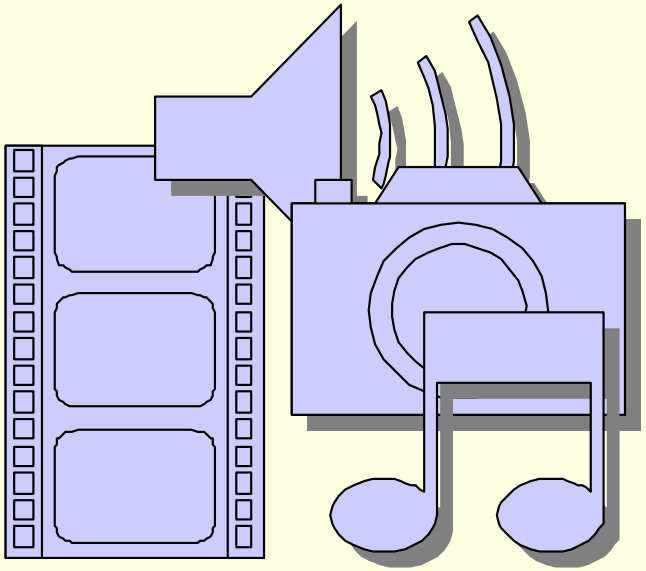
- An arrangement of letters, numbers, and special function keys that act as the primary input device to the computer.

Mouse

- An input device that allows the user to manipulate objects on the screen by moving the device along the surface of a desk.



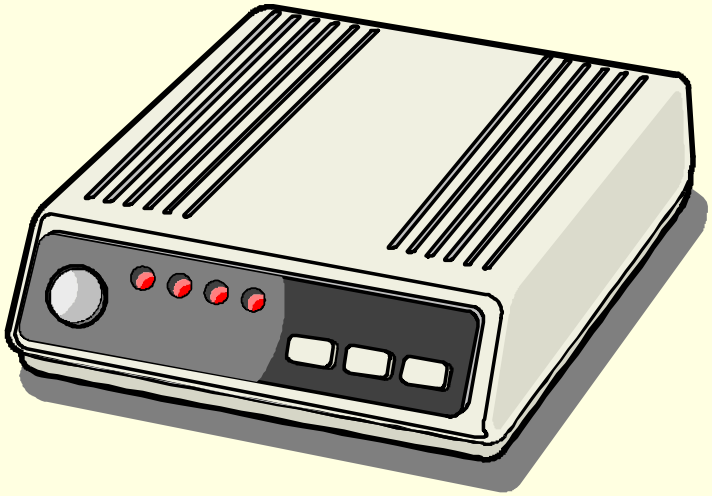
Sound Card



- A circuit board that gives the computer the ability to accept audio input, play sound files, and produce audio output through speakers or headphones.

Modem

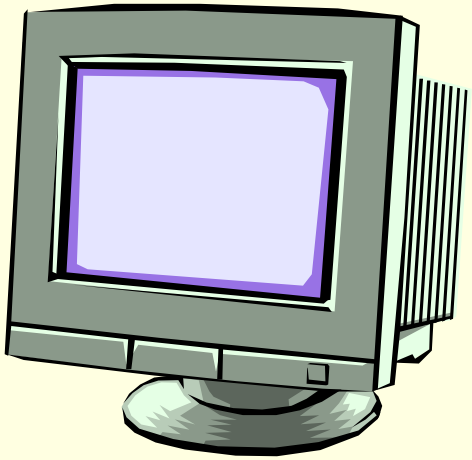
- A device that sends and receives data to and from computers over telephone lines.



Output Devices

- Devices that display, print, or transmit the results of processing from the computer's memory.

Monitor



- Display device that forms an image by converting electrical signals from the computer into points of colored light on the screen.
  - Resolution
    - The density of the grid used to display or print text and graphics; the greater the horizontal and vertical density, the higher the resolution.
  - Pixels
    - The smallest unit in a graphic image; computer display devices use a matrix of pixels to display text and graphics.

Storage Devices

- Used to keep data when the power to the computer is turned off.
- Different forms
  - Hard disk
  - Floppy or zip disks
  - CD-Writer

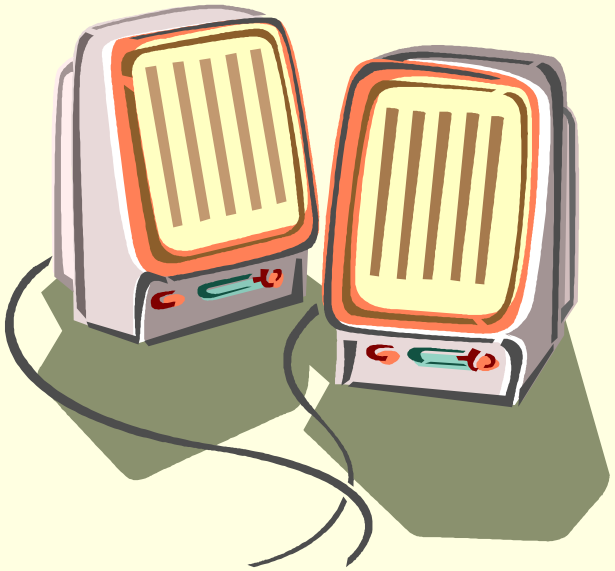


Printer

- Output device that produces text or graphical images on paper.



Speakers



- Output devices that receive signals from the computer’s sound card to play music, narration, or sound effects.